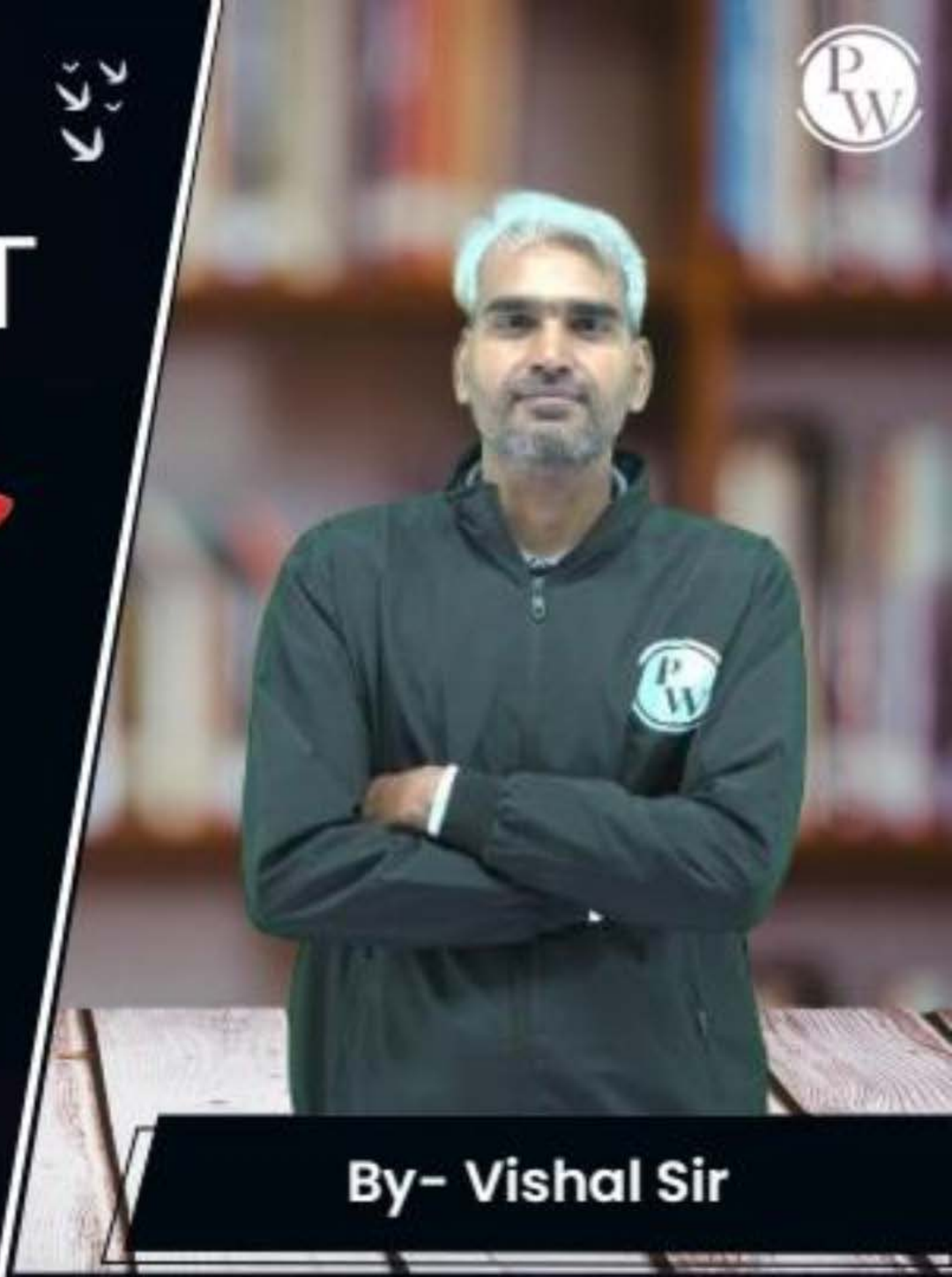
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 12



By- Vishal Sir

Recap of Previous Lecture



Topic Normal forms

Topic First normal form (1NF)

Topic Redundancy in relation because of FD

Slide

Topics to be Covered



- Topic: Second normal form (2NF)
- Topic: Third normal form (3NF)
- Topic: Boyce Codd normal form (BCNF)
- Topic: Identifying the normal form of the relation
- Topic: Decomposition of relation

Slide



Topic : Normalization

Normalization is the process of decomposing the relation into sub-relations s.t redundancy is reduced/eliminated

- Normal forms are used to describe the type of redundancy present in the relation.



Topic : Normal forms

There are various normal forms

1 NF ✓ 2 NF ✓ 3 NF ✓ BCNF ✓ | 4 NF ✓

* Upto BCNF we try to eliminate the redundancies present in the relation because of Functional dependencies

* If the relation is in BCNF then there will be 0% redundancy in the relation because of functional dependencies, but a relation which is in BCNF may suffer from redundancies present in the relation because of multi-valued dependencies

1

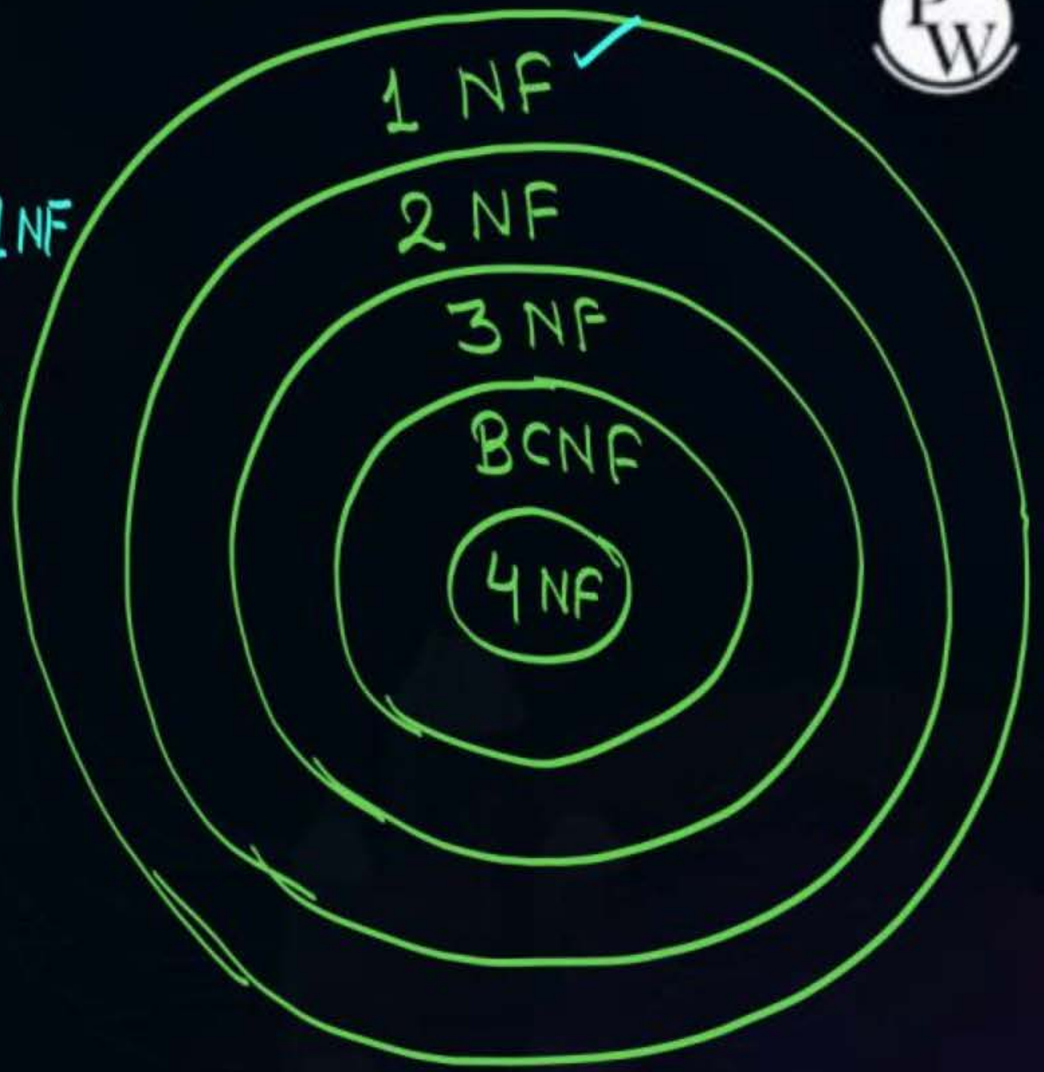
4 NF is related Multi-valued dependencies

In 4F we will try to eliminate the redundancy present in the relation because of Multi-valued dependency (MVD)



Topic : Normal forms

- If relation is in 2NF then it is also in 1NF
- If relation is in 3NF then it is also in 2NF
- If relation is in BCNF then it is also in 3NF
- If relation is in 4NF then it is also in BCNF





Topic : First normal form (1NF)

A database is in 1 NF only if it does not Contain any multi-valued attributes

Multi-valued attribute

Sid	Project
S_1	P_1, P_2
S_2	P_3, P_4
S_3	P_1

Multivalued attribute exists in the database
∴ Database is not in 1NF

Convert it into 1NF

i.e. Convert Multivalued attribute in Single-Valued attribute

Converted into single valued attribute

Sid	Project
S_1	P_1
S_1	P_2
S_2	P_3
S_2	P_4
S_3	P_1

No multivalued attribute exists., ∴ At least in 1NF



Topic : First normal form (1NF)

* In relational table no entry of the table can contain more than one value.

∴ By default normal form of relational database is 1NF.

- i.e. Every relation is at least in First normal form.



Topic : Redundancy in relation because of FD

Rule 1 :- A non-trivial functional dependency '$X \rightarrow Y$' does not cause any redundancy in the relation if 'X' is a Super key of the relation.

X → it is a SK {∵ Values will be unique}

A	B	C	Y
1	2	1	
3	1	1	
2	2	1	
3	2	1	
4	1	2	

it is not redundancy it is information



Topic : Redundancy in relation because of FD

Rule 2 :- A non-trivial functional dependency '$X \rightarrow Y$' will always cause redundancy in the relation if 'X' is not a Super key

X Y

A	B	C
1	5	-
1	5	-
1	5	-
2	2	-
3	4	-
3	4	-

it is redundancy

Note: ① In FD $X \rightarrow Y$,

if X is not a Super key, then Y can never be a superkey.

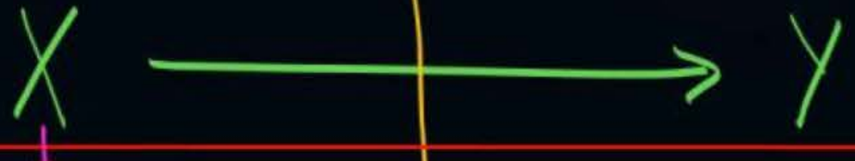
② Neither X can be a S.K (for C.K) nor Y can be a S.K (for C.K)

if $X \rightarrow y$ causes redundancy



Topic : Redundancy in relation because of FD

Possible types of Functional dependencies which may cause redundancy in the relation



① Proper subset of C.K → Non-prime attribute

② Non-prime attribute → Non prime attribute

③ Proper subset of one C.K → Proper subset of another C.K.

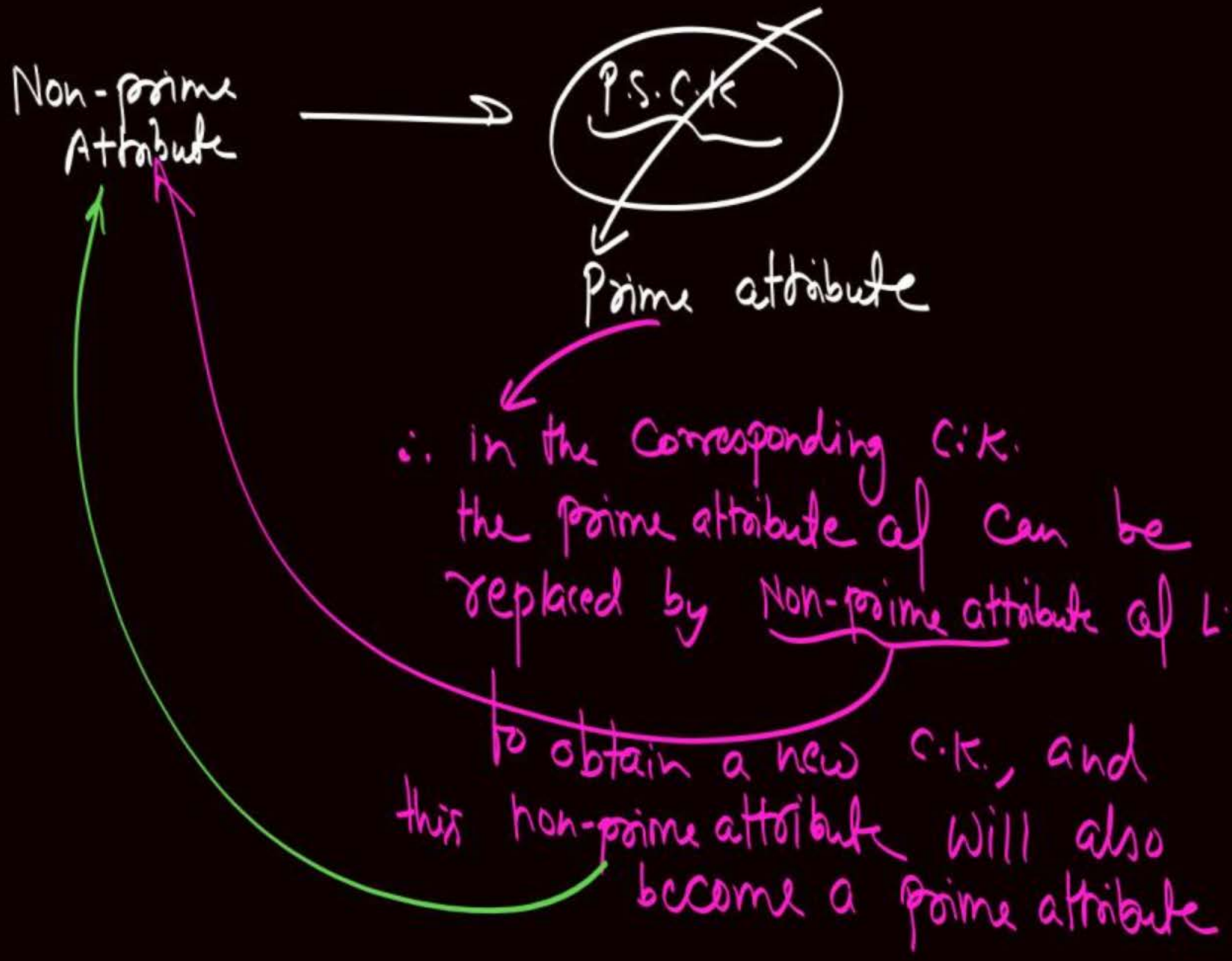


Topic : Redundancy in relation because of FD

Possible types of Functional dependencies which may cause redundancy in the relation

$X \rightarrow Y$

X		Y
① Proper subset of C.K	→	Non-prime attribute
② Non-prime attribute	→	Non prime attribute
③ Proper subset of one C.K	→	Proper subset of another C.K.
it is never Possible ④ Non-prime attribute	→	Proper subset of C.K.


Non-prime Attribute
P.S.C.K
Prime attribute
∴ in the Corresponding C.K.
the prime attribute of can be
replaced by Non-prime attribute of L.H.S.
to obtain a new C.K., and
this non-prime attribute will also
become a prime attribute



Topic : Redundancy in relation because of FD

Possible types of Functional dependencies which may cause redundancy in the relation

$X \longrightarrow Y$

① Proper subset of C.K ⟶ Non-prime attribute

② Non-prime attribute ⟶ Non prime attribute

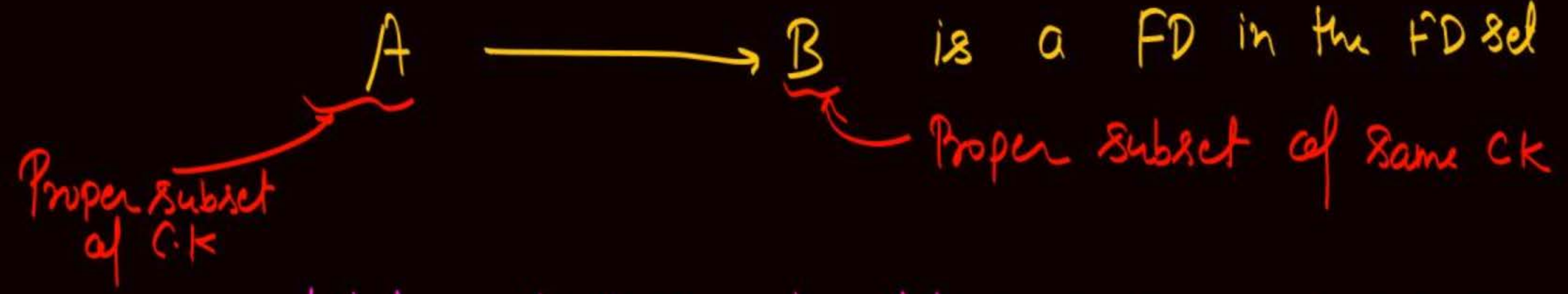
③ Proper subset of one C.K ⟶ Proper subset of another C.K.

④

⑤ Proper subset of a C.K ⟶ Proper subset of same C.K

Such FD is not possible

- Let (AB) is a candidate key and



Whatever that can be determined using attribute B can also be determined using attribute A.

∴ (AB) is not minimal because 'B' can actually be removed from the set, and hence (AB) can not be the C.K because it is not minimal

Topic : Redundancy in relation because of FD

Possible types of Functional dependencies which may cause redundancy in the relation



⑥ Proper subset of C.K → (Proper subset of another C.K + Non-prime attribute)

We can split it into two parts

P.S. C.K → P.S. of another C.K (it is our Case ③)

&

P.S. C.K → Nonprime attribute (it is our Case ①)



Topic : Redundancy in relation because of FD

Possible types of functional dependencies which may cause redundancy in the relation

$X \longrightarrow Y$

(7) (Proper Subset of a C.K + Non-prime attribut) $\longrightarrow$ Any thing may be present in R.HS.

We will focus on this type of FDs during the discussion of 3 NF.



Topic : Redundancy in relation because of FD

Possible types of Functional dependencies which may cause redundancy in the relation

$X \rightarrow Y$

IMP Case ① ① Proper subset of C.K ⟶ Non-prime attribute

Case ② ② Non-prime attribute ⟶ Non prime attribute

Case ③ ③ Proper subset of one C.K ⟶ Proper subset of another C.K.

We only need to remember these three types of FDs



Topic : Redundancy in relation because of FD

	1 NF	2 NF	3NF	BCNF
Case (1)	Allowed	Not allowed	Not allowed	Not allowed
Case (2)	Allowed	Allowed	Not allowed	Not allowed
Case (3)	Allowed	Allowed	Allowed	Not allowed



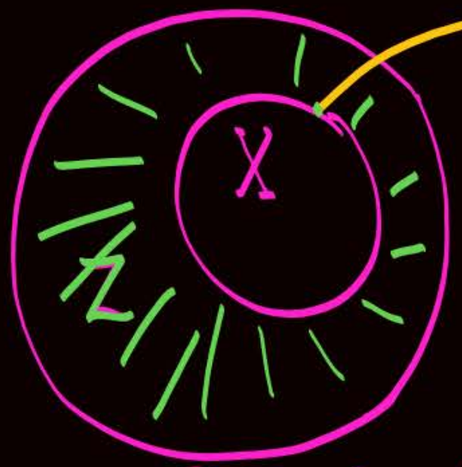
Topic : Second normal form (2NF)

A relation R is in 2 NF only if,

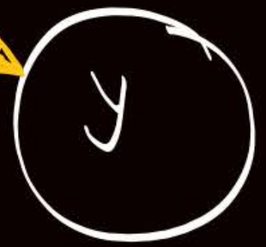
① Relation R must be in 1 NF

and ② Relation R must not contain any partial dependency.

Partial dependency :-



C.K = (XZ)



Non-prime attribute

$X \longrightarrow Y \Longrightarrow$

(Proper Subset of C.K) (Non-prime Attribute)

then

$XZ \rightarrow Y$ is a Partial dependency

- Y is not dependent on Full Candidate Key,
- A Part of Candidate Key is Capable of determining 'Y'

If y is dependent on full candidate key

i.e. $\underbrace{XZ}_{C.K} \longrightarrow Y$

then this dependency is called full dependency

If proper subset 'X' of candidate key (XZ)

can determine Y, then

$\underbrace{XZ}_{C.K} \longrightarrow Y$

is not a full dependency

it is a partial dependency

Note: If a non-trivial FD of type

Proper subset of C.K ⟶ Non-prime attribute

exists in the relation then, it ensures that

Partial dependency exists in the relation.



Topic : Third normal form (3NF)

A relation R is in 3NF only if,
every non-trivial Functional dependency
$X \longrightarrow Y$ is with

(i) X as a Super key

or (ii) Y should be a prime attribute.

If Y is a prime attribute in $X \rightarrow Y$ then Case ① & Case ② type of Functional dependencies are not allowed!



Topic : Third normal form (3NF)

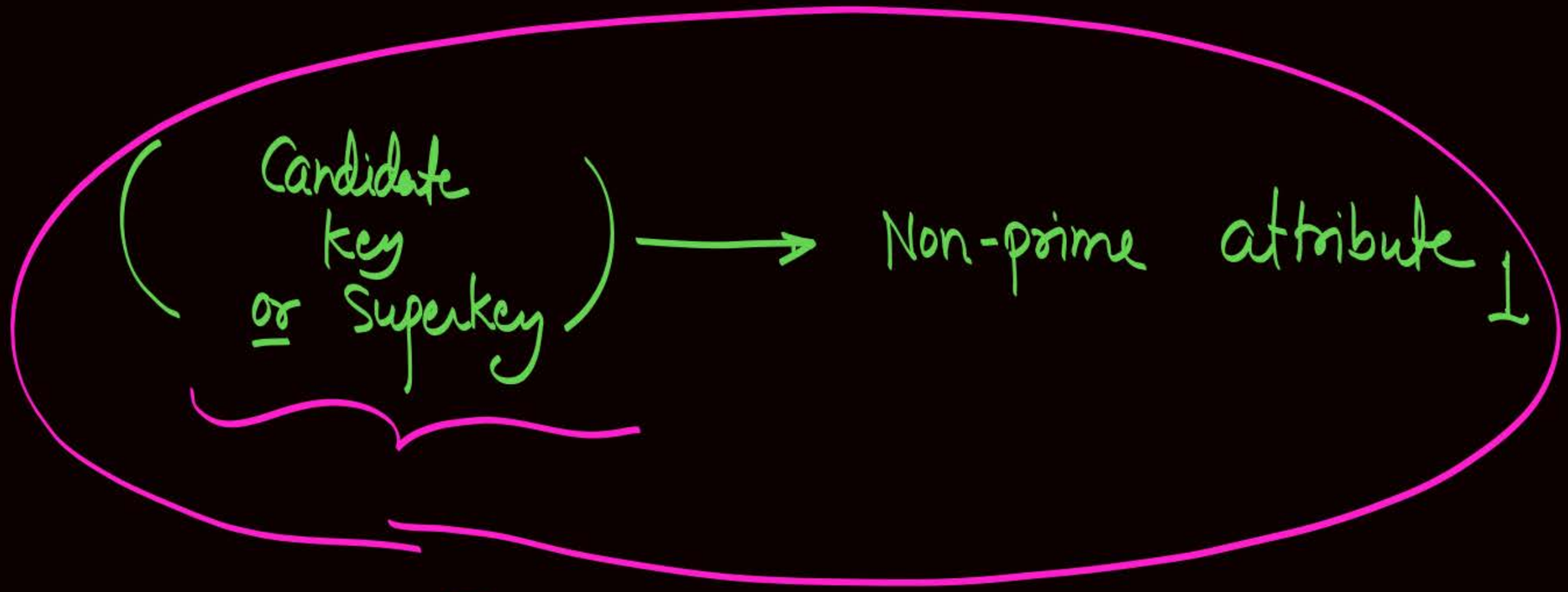
Book's Definition

A relation R is in 3 NF only if

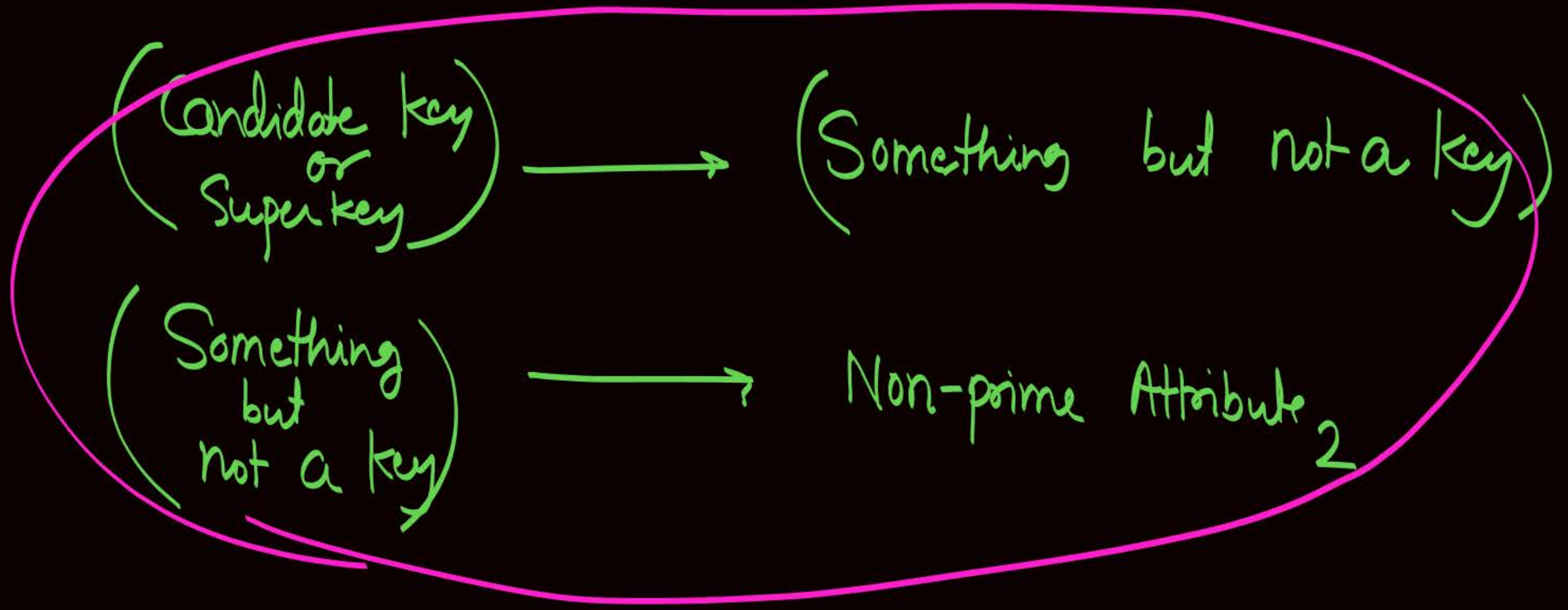
① R is in 2 NF

And ② R should not contain any transitive dependency.

A dependency is called transitive dependency if a non-prime attribute is transitively dependent on candidate key {i.e. Not directly dependent on CK}



In this case Non-prime Attribute 1 is directly dependent on key



In this case Non-prime attribute 2 is transitively dependent on key



Topic : Boyce Codd normal form (BCNF)

A relation R is in BCNF only if every non-trivial FD "$X \rightarrow Y$" is with "X" as a Superkey.

Topic: Identifying the normal form of the Relation:

Q: Consider the relational schema R(A, B, C, D, E, F) with following FD set F, Identify the normal form of relation

F = {
- $AB \rightarrow CD$ { Allowed upto BCNF } — AB: S.K
- $D \rightarrow A$ { Allowed upto 3NF but not allowed in BCNF } — Case (iii) — D: Proper subset of C.K (DB); A: Proper subset of C.K (AB)
- $C \rightarrow E$ { Allowed in 2NF, but not allowed in 3NF } — Case (ii) — C: Non-prime Attribute; E: Non-prime Attribute
- $D \rightarrow F$ { Allowed in 1 NF, Not allowed in 2NF } — Case (i) — D: Proper Subset of C.K; F: Non-prime Attribute
}

Step 1: Identify all C.Ks of relation



AB is a C.K
Prime Attributes = {A, B}
D → A
∴ Replace A by D
DB is a C.K.
Prime Attributes = {A, B, D}

(AB) & (DB) are the only C.Ks
∴ Prime Attributes = {A, B, D}
Non-Prime Attributes = {C, E, F}

Highest Normal form w.r.t.

$AB \rightarrow CD$ is BCNF

$D \rightarrow A$ is 3NF

$C \rightarrow E$ is 2NF

$D \rightarrow F$ is 1NF

Normal form of the relation will be least normal form forced by any FD.

In the above example least Normal form is 1NF.

∴ given relation is in 1NF but not in 2NF

* Identify the normal form of the relation and decompose the relation to improve the normal form :→

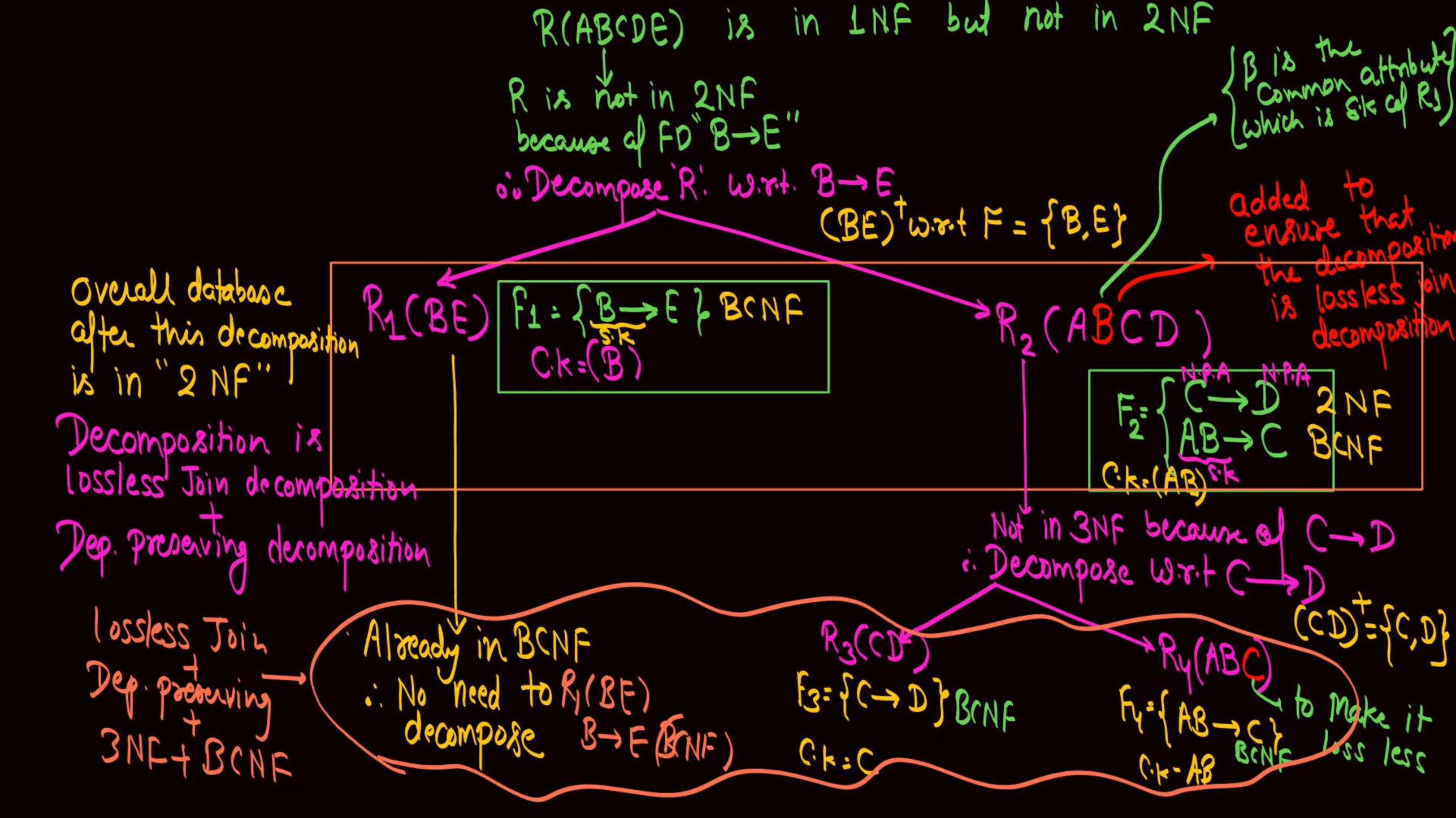
Q: Consider the relational schema R(A,B,C,D,E) with FD set

C.k = (AB) is the only C.K.

$$F = \begin{cases} \underbrace{AB}_{S.K} \longrightarrow C & \{BCNF\} \\ \underbrace{C}_{N.P.A} \longrightarrow \underbrace{D}_{N.P.A} & \{2NF\} \\ \underbrace{B}_{P.S.CK} \longrightarrow \underbrace{E}_{N.P.A} & \{1NF\} \end{cases}$$

∴ Rel^{h} R is in 1NF

Identify the normal form of the relation, and decompose the relation upto BCNF


R(ABCDE) is in 1NF but not in 2NF
R is not in 2NF because of FD "B→E"
∴ Decompose R: w.r.t. B→E
(BE)+ w.r.t F = {B,E}
B is the common attribute which is b/w of R1
added to ensure that the decomposition is lossless join decomposition
R1(BE)
F1 = {B→E} BCNF
C.k = (B)
R2(ABCD)
F2 = {C→D N.P.A, AB→C N.P.A} 2NF BCNF
C.k = (AB)
Overall database after this decomposition is in "2 NF"
Decomposition is lossless Join decomposition + Dep. preserving decomposition
Not in 3NF because of C→D
∴ Decompose w.r.t C→D
(CD)+ = {C,D}
lossless Join + Dep. preserving + 3NF + BCNF
Already in BCNF
∴ No need to R1(BE) decompose B→E (BCNF)
R3(CD)
F3 = {C→D} BCNF
C.k = C
R4(ABC)
F4 = {AB→C} BCNF
C.k = AB
to make it loss less



2 mins Summary

- **Topic** Second normal form (2NF)
- **Topic** Third normal form (3NF)
- **Topic** Boyce Codd normal form (BCNF)
- **Topic** Identifying the normal form of the relation
- **Topic** Decomposition of relation

THANK - YOU